1. Who is going to benefit the most from your business intelligence solution at the end of the semester? (Yanit) + introduction

- As organizations continue to collect and process large amounts of data, many still lack the tools needed to turn that data into useful and actionable insights. The Division of Purchasing and Supply (DPS) is one such organization. With just over 100 employees, DPS processes a high volume of procurement data each year, including a significant amount from agencies such as the Virginia Department of Corrections (VADOC). Manually analyzing this procurement data can be challenging due to the large number of records and the time required to review spending on a daily or weekly basis. To address this challenge, we propose a Business Intelligence (BI) solution that will include a centralized repository of procurement data, along with dashboards and predictive analytics tools. This solution will support deeper analysis of spending patterns, improve reporting capabilities, and help guide more informed decisions.
- With the successful implementation of the BI solution, several stakeholder groups will benefit; however, the primary users will be DPS procurement analysts and executive leadership, along with stakeholders at VADOC. DPS analysts will have a centralized location where all procurement data is regularly integrated and updated, making it easier to analyze spending trends, create reports, and share insights with leadership. As a result, analysts will spend less time manually preparing data and more time focusing on understanding patterns and identifying opportunities for improvement.
- The dashboards created for DPS leaders and decision-makers will provide clear views of trends and future spending patterns, helping support better budgeting, cost management, and strategic planning. VADOC stakeholders and other affected groups will also gain greater visibility into procurement activity, supporting daily operations and resource planning. Overall, the BI solution will improve efficiency, transparency, and the effective use of data across DPS and VADOC.

2. Why is your solution going to be the most optimal? Remember to include "why your team" and not focus only technology**. Faustina**

Our solution is optimal because it focuses on producing reliable, structured, and actionable insights from a complex dataset while remaining achievable within a semester-long student project. By using industry-standard BI tools and proven data practices, we ensure the system is realistic, scalable, and aligned with how organizations actually use analytics. At the same time, the sprint-based approach allows us to test ideas, improve our models, and respond to feedback as we learn more about the data.

3. What kinds of resistance might you encounter as you implement this solution? What can you recommend now as a way to get your customer and your team receptive to the solution? **Yabby**

The proposed Business Intelligence solution will face implementation challenges because of employee resistance to change and technology adoption and their doubts about data trustworthiness. Employees may feel comfortable using existing manual reports and may hesitate to adopt a new self-service dashboard environment. Some stakeholders may question the accuracy of transformed data or predictive analytics, especially for high-visibility compliance reports such as SWAM spend. Security controls and access permissions of the centralized repository system will raise concerns among users. Users believe that they will face extra work from learning the new visualization tool instead of achieving improved efficiency. The project will experience delays because of these two factors which require immediate solution during the initial project stages.

The customer and our team need to stay open to work because we need to use an agile approach together with our implementation plan which follows the RFP document's sprint-based delivery system. The DPS stakeholders who matter most will assist us in selecting tools and designing dashboards to create a solution that meets actual business demands. The first sprints will deliver high-value outputs which show immediate benefits through total spend reports and compliance dashboards. The organization will conduct training through organized sessions while providing users with documentation to help them become confident and handle technical challenges. The organization will establish trust in its centralized repository through its transparent data validation processes. The combination of stakeholder engagement with incremental value creation and effective communication will result in better product adoption together with sustained operational success.

4. Will this be easy or hard for the customer to use going forward? Why? Will they need special tools, training? (Breanna)

The BI solution will be easy for the customer to use going forward because it is built with tools that DPS already works with, such as Excel and Power BI. Most of the complex setup happens during development, so the customer will only need to refresh the data and use the dashboards through simple filters and visuals. They will not need special tools or advanced technical skills. A short training session will be enough to show them how to navigate the dashboards and understand the key insights. By keeping everything familiar and straightforward, the customer can continue using the system confidently in the future.

5. A financial writeup is out of scope for this proposal in terms of details, but please include if this is expensive or inexpensive, what elements make it so.(Beth)

The solution is **relatively inexpensive** because it leverages cloud-based BI tools and existing data infrastructure. Costs remain low due to minimal hardware requirements and scalable licensing models.

The most expensive elements are:

- Data cleaning and integration effort
- ETL automation setup
- User training and change-management support

The most inexpensive elements are:

- Dashboard creation once data pipelines are established
- Cloud storage scalability
- Maintenance due to automation reducing manual labor

6. Include in your business proposal an analysis and selection by the team for the tool or tools you intend to use to create this value. **Show your business decision analysis method and thinking** that supports your decision. For example, use a Pugh matrix or any other type of repeatable decision analysis method. (Beth)

Tool Analysis & Selection — Tableau Chosen

- The team evaluated four BI tools: **Tableau, Power BI, Google Looker Studio, and Qlik Sense**.
- A **Pugh decision matrix** was used to compare tools across key criteria: cost, scalability, security, ease of integration, user experience, and learning curve.
- Tableau scored highest in **visualization quality**, **interactivity**, and **executive-level storytelling**, which were the most important factors for DPS/DOC reporting needs.
- Tableau provides strong performance with **large and complex datasets**, making it ideal for statewide procurement and compliance data.
- The platform offers an intuitive **drag-and-drop interface**, reducing training time and making it accessible for analysts and managers.
- Tableau integrates smoothly with multiple data sources, supporting the project's goal of **centralized, automated reporting**.
- Although Tableau can be more expensive than some alternatives, its **advanced analytics**, **visual flexibility**, and **long-term value** justify the investment.
- The team selected Tableau because it best aligns with the project's goals of **clarity, automation, trend analysis, and executive decision support**.

Pugh Matrix Summary (Tableau Selected)

- **Criteria Used:**
    - Cost
    - Scalability

    - Security
    - Ease of Integration
    - User Experience
    - Learning Curve
- **Weighted Scoring Outcome:**
    - **Tableau ranked highest overall**, driven by superior visualization capabilities and strong integration options.
    - Power BI ranked second, performing well in cost and integration but lower in visualization flexibility.
    - Looker and Qlik scored lower due to limited features, higher complexity, or weaker visualization capabilities.
- **Final Decision:**
    - **Tableau selected** as the primary BI tool because it provides the strongest combination of usability, visual storytelling, and analytical depth required for DPS/DOC procurement and compliance reporting.

Decision Criteria:

- Cost
- Scalability
- Security
- Ease of Integration
- User Experience
- Learning Curve

Include your full team **TOPs document**.

Our Epics

Yanit's 2 epics:

- Business analyst:
  - As a data analyst, I want a centralized place where all VADOC procurement data is stored and updated, so I can easily analyze spending trends, catch unusual changes, and better support decision-making.
- Shareholder:
  - As DPS, I want to use predictive models and dashboards to see patterns and trends in procurement data, so we can estimate future spending, plan budgets better, and make more informed decisions.

Yabby's 2 epics:

1. The DPS leadership team together with procurement analysts require a secure central business intelligence system which will convert their procurement information into standardized dashboards and fiscal year financial reports for their operational needs. The epic delivers benefits through its development of a reliable main reference point which enhances organizational visibility while decreasing manual reporting needs and improving cost management operations throughout the agency.
2. The DPS executives together with the compliance stakeholders require automated SWAM compliance reporting together with predictive purchasing analytics to monitor trends and maintain regulatory compliance and enhance their procurement operations. The epic enables organizations to move from traditional reporting methods to forward-looking strategic forecasting which enhances their ability to manage operational activities.

Faustina's 2 epics:

- Compliance & Trust:

As DPS, we need a clear and trusted view of procurement compliance so they can meet regulations, ensure fairness, and report accurately to state leadership. This epic focuses on creating automated compliance dashboards, especially for SWAM spend, so users can instantly see whether purchasing goals are being met. By reducing manual reporting and increasing transparency, this epic builds trust in the data and supports ethical, accountable government purchasing.

- Smarter Spending Decisions:

As data analysts, we need to understand trends and patterns in spending so they can reduce waste, improve vendor selection, and plan for future demand. This epic focuses on comparative reports, category analysis, and predictive modeling to identify where money is being spent and where it could be spent better.

Beth's 2 epics:

Automated Procurement & Compliance Reporting: This semester, our team will create an automated reporting system that eliminates manual data pulls and provides procurement managers with real-time visibility into vendor spend, SWAM compliance, and purchasing trends. By integrating data sources and designing intuitive Tableau dashboards, this epic focuses on reducing human workload, improving accuracy, and enabling faster decision-making. The value created will directly support analysts and managers who currently spend hours compiling reports, allowing them to shift their time toward strategic analysis rather than repetitive tasks.

Centralized Data Integration & Executive Trend Analysis : Our second epic centers on building a unified BI environment that consolidates procurement data across all DOC facilities, giving executives a single, reliable source of truth for long-term planning and anomaly detection. This epic emphasizes both the technical challenge of integrating multiple systems and the human need for clear, actionable insights that support leadership decisions. Through Tableau's advanced visualization capabilities, we will deliver interactive trend analysis tools that help executives identify patterns, forecast needs, and strengthen statewide procurement oversight.

Breanna 2 epics:

As a DPS analyst, I need a dashboard that is simple to navigate and easy to understand, so I can quickly find the information I need without relying on complicated tools or extra training. This matters because the system should feel natural to use long-term, even for people who are not technical, and it should help analysts work faster instead of slowing them down.

As a member of the project team, I need to design the BI solution in a way that requires minimal training and uses tools DPS already knows, so the customer can maintain it easily after the semester ends. By keeping the process straightforward and avoiding unnecessary complexity, the system will be sustainable, low-maintenance, and comfortable for the customer to use going forward.

COMBINED Epic Stories:

Epic Story #1 – Clarity in Reporting and Use of Data

As DPS analysts and members of executive leadership at VADOC, we want all procurement data to be stored in one centralized source and updated regularly so we can easily analyze spending, create reports, and reduce the time spent manually working with data.

Having this centralized system will allow us to track trends, improve reporting accuracy, meet compliance requirements (such as SWAM), and provide a clear and reliable view of procurement activity.

Epic Story #2 – Smarter Decision-Making and Strategic Planning

As DPS executives and stakeholders, we want access to easy-to-use dashboards and predictive tools that show patterns and trends in procurement data so we can make better budgeting, cost management, compliance, and spending decisions moving forward.

These tools will allow leadership to quickly understand current spending, identify unusual changes, and plan for future needs instead of only reviewing past data.

Our team has created two epic stories to guide the development of the BI solution and explain the main value the system will provide. The first epic focuses on improving reporting and making procurement data easier to use. DPS analysts and executive leadership need all procurement data from across the state to be stored in one central place that is reliable and regularly updated. This will allow them to analyze spending, create accurate reports, and spend less time working with data manually. Having one central source of information will also help support compliance needs such as SWAM reporting and give a clear view of procurement activity across the organization.

The second epic focuses on helping decision-makers make better decisions and plan for the future. DPS executives and stakeholders need simple dashboards and predictive tools that show trends and patterns in procurement data. These tools will help leaders understand how money is being spent now, notice any unusual changes, and predict future needs. With a clearer view of procurement data, decision-makers will be able to plan budgets better, manage costs, stay compliant with regulations, and make smarter long-term decisions instead of only relying on past reports.